
Project Plan

Entanglement Swapping
in Quantum Repeater Chains

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May 22, 2026

Stack: Julia · QuantumSavory.jl · ConcurrentSim.jl

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1 Objective

Simulate a linear quantum network composed of two terminal nodes (Alice and Bob) and N intermediate quantum repeaters. The goal is to establish end-to-end entanglement via the *entanglement swapping* protocol, analysing performance in both ideal and non-ideal scenarios.

The implementation must be written in **Julia** using the **QuantumSavory.jl** framework.

2 Phase 0 — Environment Setup

Objective: prepare the development environment and become familiar with the framework.

1. Install Julia (v1.10+) and initialise a project environment.
2. Add dependencies to `Project.toml`:
 - ▶ `QuantumSavory.jl` — quantum network simulation framework
 - ▶ `ConcurrentSim.jl` — discrete-event simulation (QuantumSavory dependency)
 - ▶ `QuantumOpticsBase.jl` / `QuantumClifford.jl` — backends for quantum states
 - ▶ `Plots.jl` or `CairoMakie.jl` — plot generation
 - ▶ `Statistics.jl`, `Distributions.jl` — statistical analysis
3. Study the QuantumSavory documentation and examples related to repeater chains and entanglement swapping.

3 Phase 1 — Ideal Case

Objective: verify the correctness of the implementation in the absence of noise.

3.1 Network Model

Create a linear `RegisterNet` with $N + 2$ nodes:

- ▶ **Alice and Bob** (terminal nodes): 1 quantum memory slot each, single-qubit operations.
- ▶ **Repeaters** $R_1, \dots, R_N$: 2 memory slots each, two-qubit operations (including BSM).

3.2 Bell Pair Generation

On every adjacent link $(i, i + 1)$, generate the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ perfectly and instantaneously.

3.3 Entanglement Swapping Protocol

For each repeater R_k ($k = 1, \dots, N$):

- (a) Perform a *Bell-State Measurement* (BSM) on the 2 local qubits.
- (b) Communicate the classical outcome (2 bits) to the next node.

- (c) Apply Pauli corrections (X, Z) conditioned on the outcome.

3.4 Validation

- ▶ Verify $F = \langle \Phi^+ | \rho_{AB} | \Phi^+ \rangle = 1.0$ for every N .
- ▶ Test for $N = 1, 2, 3, 5$.
- ▶ Distribution time = 0 (instantaneous generation).

$F = 1.0$ exactly for every N in the ideal case. Any deviation indicates a bug in the implementation.

4 Phase 2 — Non-Ideal Model

Objective: introduce realistic noise sources and study their effect.

4.1 Memory Noise: Depolarising Channel

Each memory slot is subject to depolarising noise with parameter p_w :

$$\mathcal{E}(\rho) = (1 - p_w) \rho + p_w \cdot \frac{I}{2} \quad (1)$$

Noise accumulates over time: the longer a qubit remains in memory, the greater the degradation. In QuantumSavory this is modelled via `Depolarization(rate)` in the register traits.

4.2 Probabilistic Bell Pair Generation

Each generation attempt succeeds with probability p_{success} and fails with probability $1 - p_{\text{success}}$. Each attempt takes one discrete time unit. The number of attempts required to generate a Bell pair follows a *geometric distribution*:

$$\Pr(T_{\text{link}} = k) = (1 - p_{\text{success}})^{k-1} p_{\text{success}}, \quad \mathbb{E}[T_{\text{link}}] = \frac{1}{p_{\text{success}}} \quad (2)$$

4.3 Protocol with Retry

1. Adjacent links attempt Bell pair generation **in parallel** and independently.
2. A link that completes before the others leaves its qubit in memory, where it undergoes decoherence.
3. Each repeater waits until both its links are ready, then performs a BSM.
4. The total time is dominated by the slowest link:

$$T_{\text{total}} = \max(T_{\text{link},1}, T_{\text{link},2}, \dots, T_{\text{link},N+1}) \quad (3)$$

The interplay between p_{success} (waiting time) and p_w (decoherence during the wait) is the **fundamental trade-off** of the system: a low generation probability increases the time spent in memory and therefore the fidelity degradation.

5 Phase 3 — Monte Carlo Simulation

Objective: collect robust statistics on fidelity and distribution time.

For each parameter combination, run M independent trials ($M = 500\text{--}1000$) and collect statistics (mean, standard deviation, confidence intervals).

5.1 Metrics

► **Fidelity:**

$$F = \langle \Phi^+ | \rho_{AB} | \Phi^+ \rangle \quad (4)$$

where ρ_{AB} is the final state of Alice's and Bob's qubits.

► **Distribution time:** number of time steps from the request to the completion of the full swapping chain.

5.2 Parameter Space

Parameter	Range	Notes
p_{success}	0.1, 0.2, ..., 1.0	10 equally spaced points
p_w	0.01, 0.05, 0.10	3 values
N	1, 3, 5	3 values

Table 1: Simulation parameters.

6 Phase 4 — Required Plots

Objective: produce the plots required by the assignment and extra plots for the 30L grade.

6.1 Plot 1: Distribution Time vs. p_{success}

- x -axis: $p_{\text{success}} \in (0, 1]$.
- y -axis: mean distribution time (time steps).
- 3 curves: $N = 1$, $N = 3$, $N = 5$.
- Add error bars (standard deviation or 95% CI).

- **Expected result:** T decreases as $\sim 1/p_{\text{success}}$; grows with N according to

$$\mathbb{E}[T_{\text{total}}] \approx \frac{H_{N+1}}{p_{\text{success}}} \quad (5)$$

where $H_n = \sum_{k=1}^n 1/k$ is the harmonic number (from the maximum of $N+1$ i.i.d. geometric variables).

6.2 Plot 2: Fidelity vs. p_{success}

- x -axis: $p_{\text{success}} \in (0, 1]$.
- y -axis: mean fidelity.
- 3 curves: $p_w = 0.01$, $p_w = 0.05$, $p_w = 0.10$.
- N fixed (e.g. $N = 3$).
- **Expected result:** F increases with p_{success} (less waiting $\Rightarrow$ less decoherence); F decreases with p_w ; for $p_{\text{success}} \rightarrow 1$ and small p_w , $F \rightarrow 1$.

6.3 Extra Plots (Optional, Recommended)

- ★ Fidelity vs. N at fixed p_{success} and p_w — highlights the scaling with chain length.
- ★ 2D heatmap: $F(p_{\text{success}}, p_w)$ for fixed N .
- ★ Comparison of numerical curves against analytical formulas.

7 Phase 5 — Analysis and Discussion

Objective: interpret the results with physical rigour and validate against theory.

1. **Scaling of time with N :** The total time is the maximum of $N+1$ independent geometric random variables. On average, $\mathbb{E}[T_{\text{total}}] \sim H_{N+1}/p_{\text{success}}$, with logarithmic growth in N .
2. **Fidelity degradation:** Each qubit undergoes depolarisation for a time T . The fidelity decays approximately as

$$F \sim \left(1 - \frac{4}{3}p_w\right)^{T \cdot N} \quad (6)$$

More repeaters $\Rightarrow$ more swap operations $\Rightarrow$ cumulative degradation.

3. **Trade-off p_{success} vs. p_w :** Low p_{success} $\Rightarrow$ high waiting time $\Rightarrow$ more decoherence $\Rightarrow$ low fidelity. In practice, repeaters with high p_{success} and long-coherence memories are required.
4. **Comparison with theory:** Validate numerical results against analytical formulas where possible (the $N = 1$ case admits a closed-form expression).
5. **Emergent effects:** Discuss any additional trends observed in the simulations (e.g. variance of the time, distribution of fidelity, etc.).

8 Phase 6 — Presentation

Objective: prepare clear and concise slides for a presentation of at most 15 minutes.

Slide	Content	Time
1	Title and context	1 min
2–3	Background: Bell pairs, entanglement swapping, repeaters	3 min
4–5	Model: network, ideal case, non-ideal parameters	2 min
6	Tech stack: Julia + QuantumSavory.jl, workflow	1 min
7	Ideal-case results ($F = 1$, validation)	1 min
8–9	Plot 1: Distribution time vs. p_{success}	2 min
10–11	Plot 2: Fidelity vs. p_{success}	2 min
12	Extra results / heatmap	1 min
13	Discussion: scaling, trade-off, analytical comparison	1.5 min
14	Conclusions and possible extensions	0.5 min

Table 2: Presentation structure.

9 Deliverables

1. **Julia source code:** simulation, analysis, plot generation.
2. **Presentation slides:** overview, workflow, results, conclusions.